Week 10
1. Databases: An Introduction

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COMP.2800 – Software Development [2026W]

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Agenda

- Preface
- The DIKW Pyramid
- Understanding Data
- Organizing Data in Files
- Problems in File Environments
- About Databases
- Types of Databases
- RDBMS
- Wrap-up

Part I
Preface

Preface

Some facts …

- 1
  - An integer
- “Course”
  - A string
- 1.27
  - A decimal number

Some more facts …

- $1*e^{20}$
  - A real number (scientific notation)
- Audio / Video
  - BLOB (Binary Large Objects)
- Web pages
  - Unstructured / Semi-structured

What’s missing

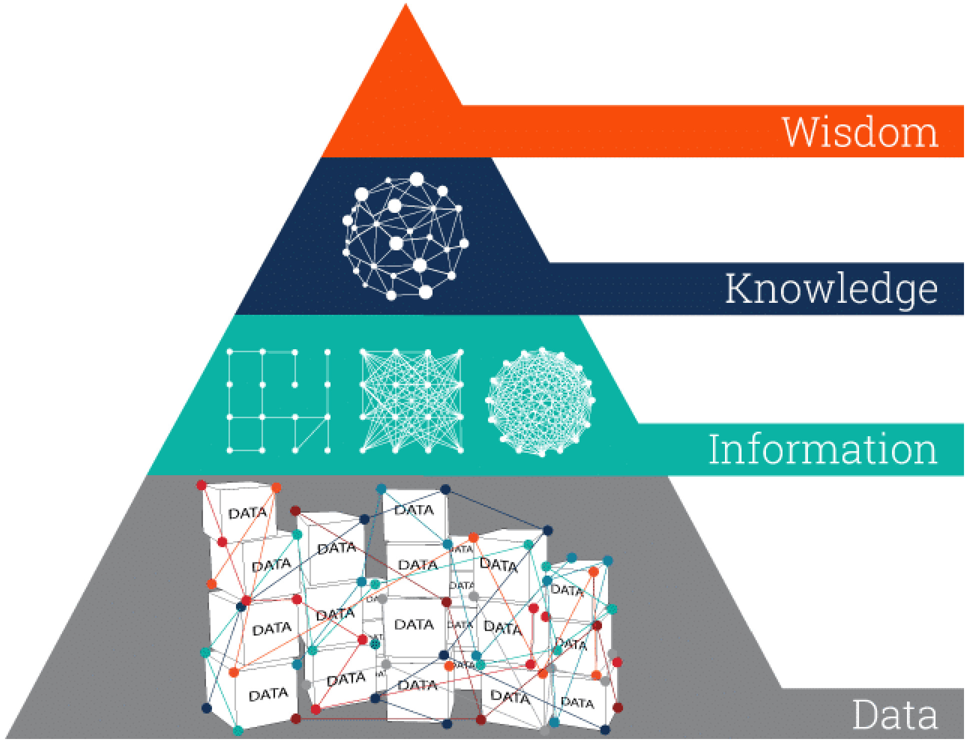
- Information
- Intelligence → Decision Making
- Knowledge → Informed Action
- Wisdom

Part II
Data – Information – Knowledge – Wisdom (DIKW)

The DIKW Pyramid

The DIKW Pyramid

- We can use the DIKW Pyramid model to explain the structural and/or functional relationships between data and information.



Each step up the pyramid answers questions about and adds value to the initial data.

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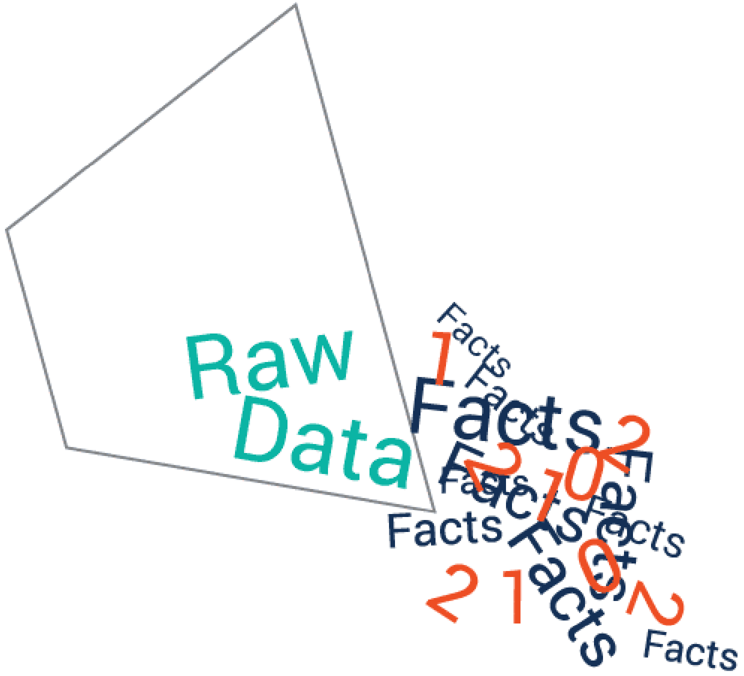
Data

- **Data**: Values without meaning.

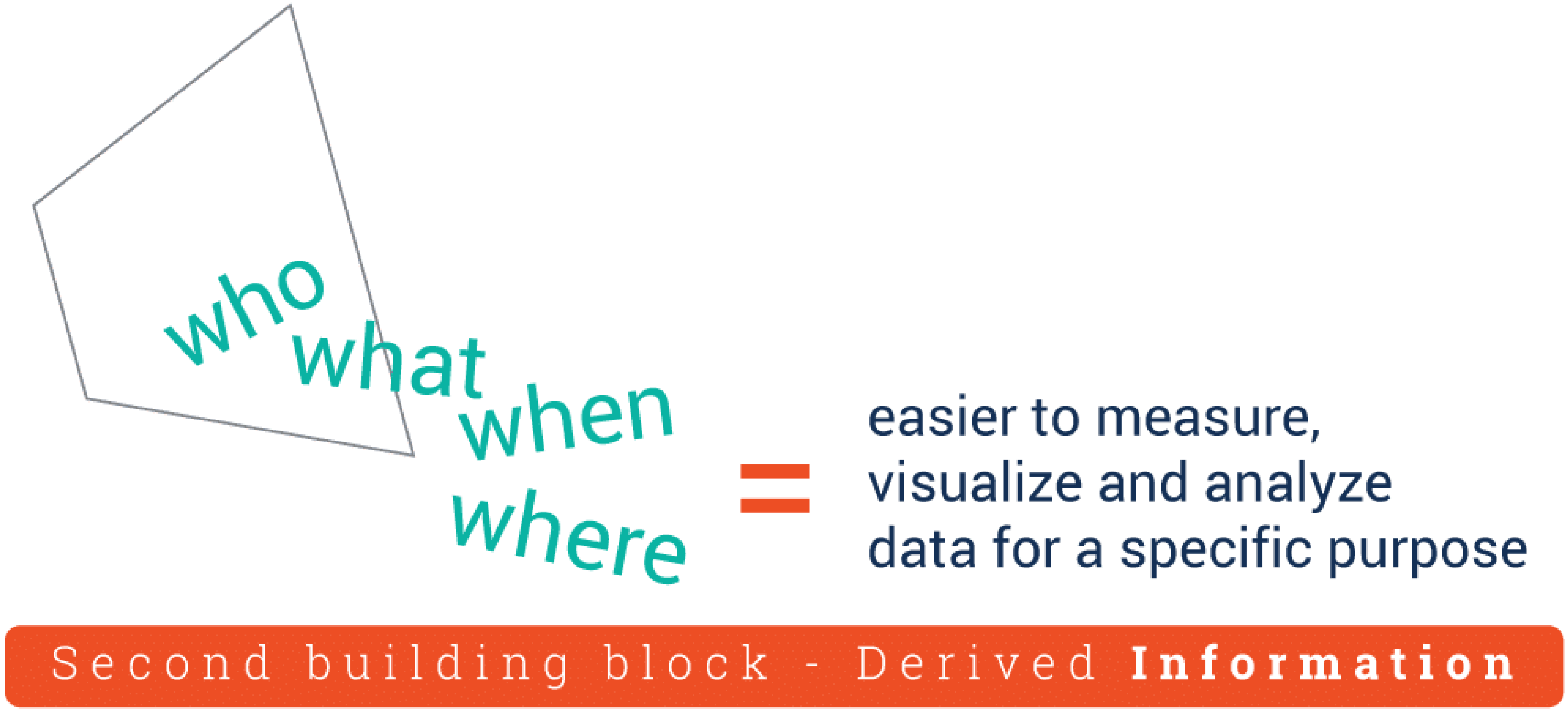


= a collection of facts in a raw or unorganized form

Base building block - Raw Data

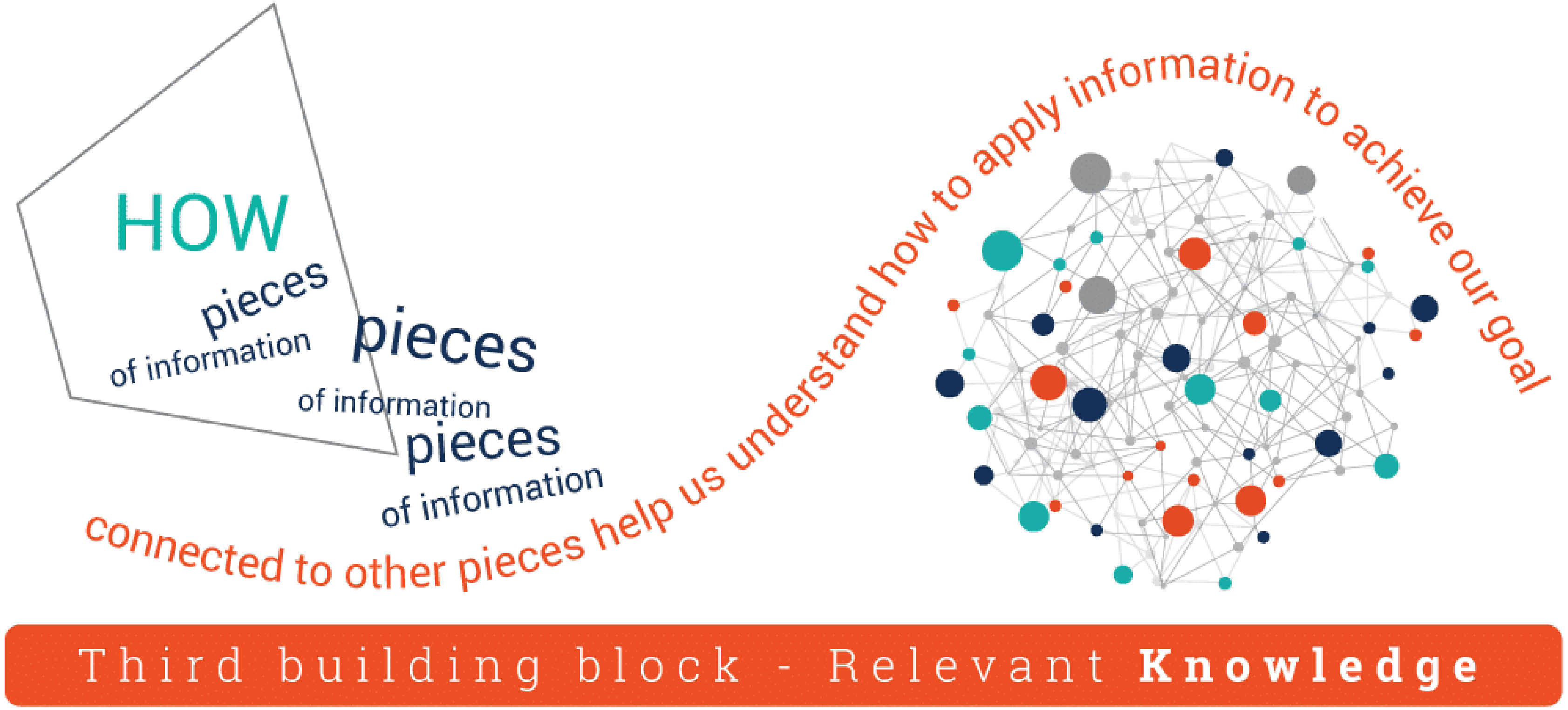
Information

- **Information**: Data with context.



Knowledge

- **Knowledge**: Information with experience.



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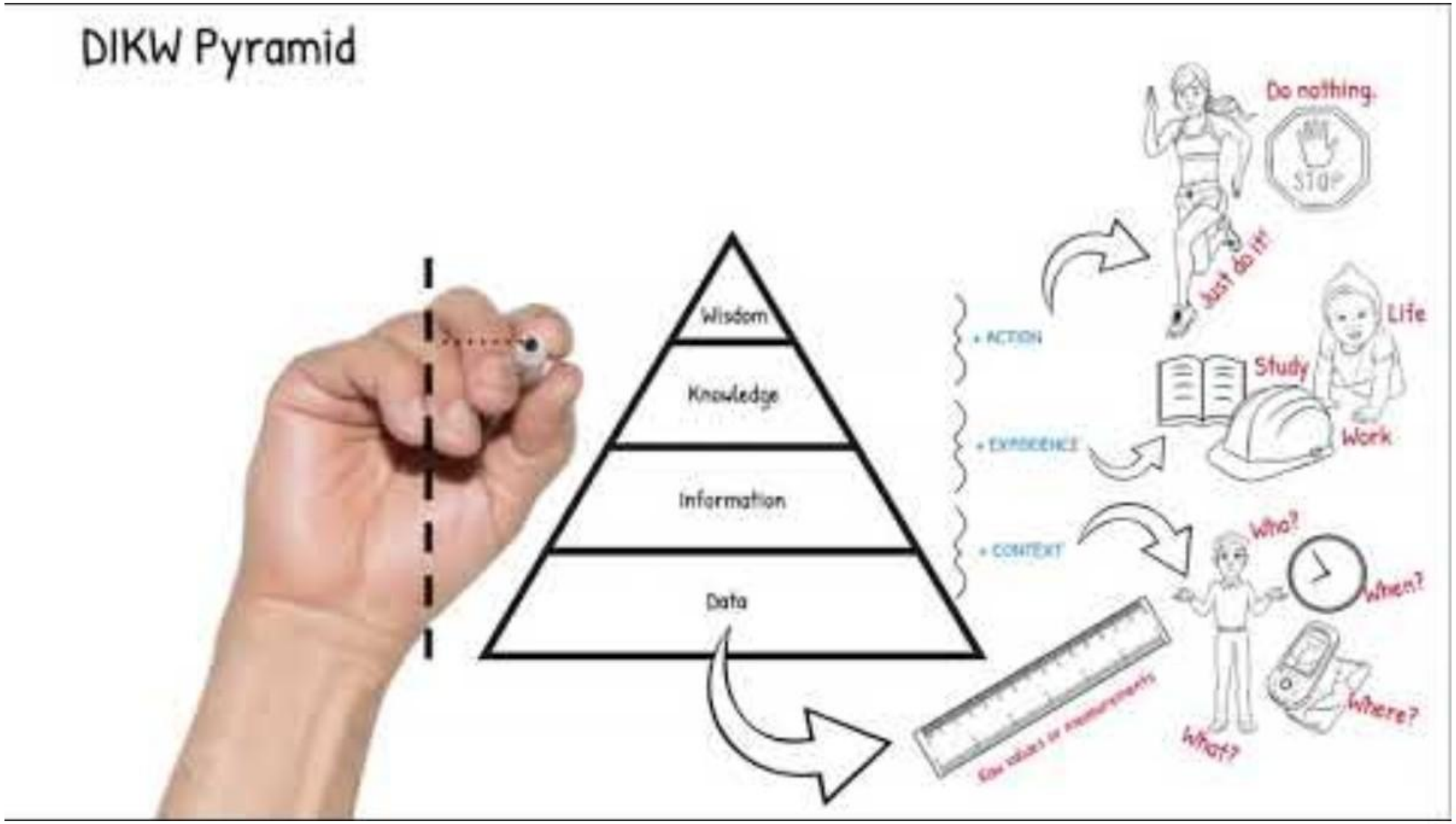
Wisdom

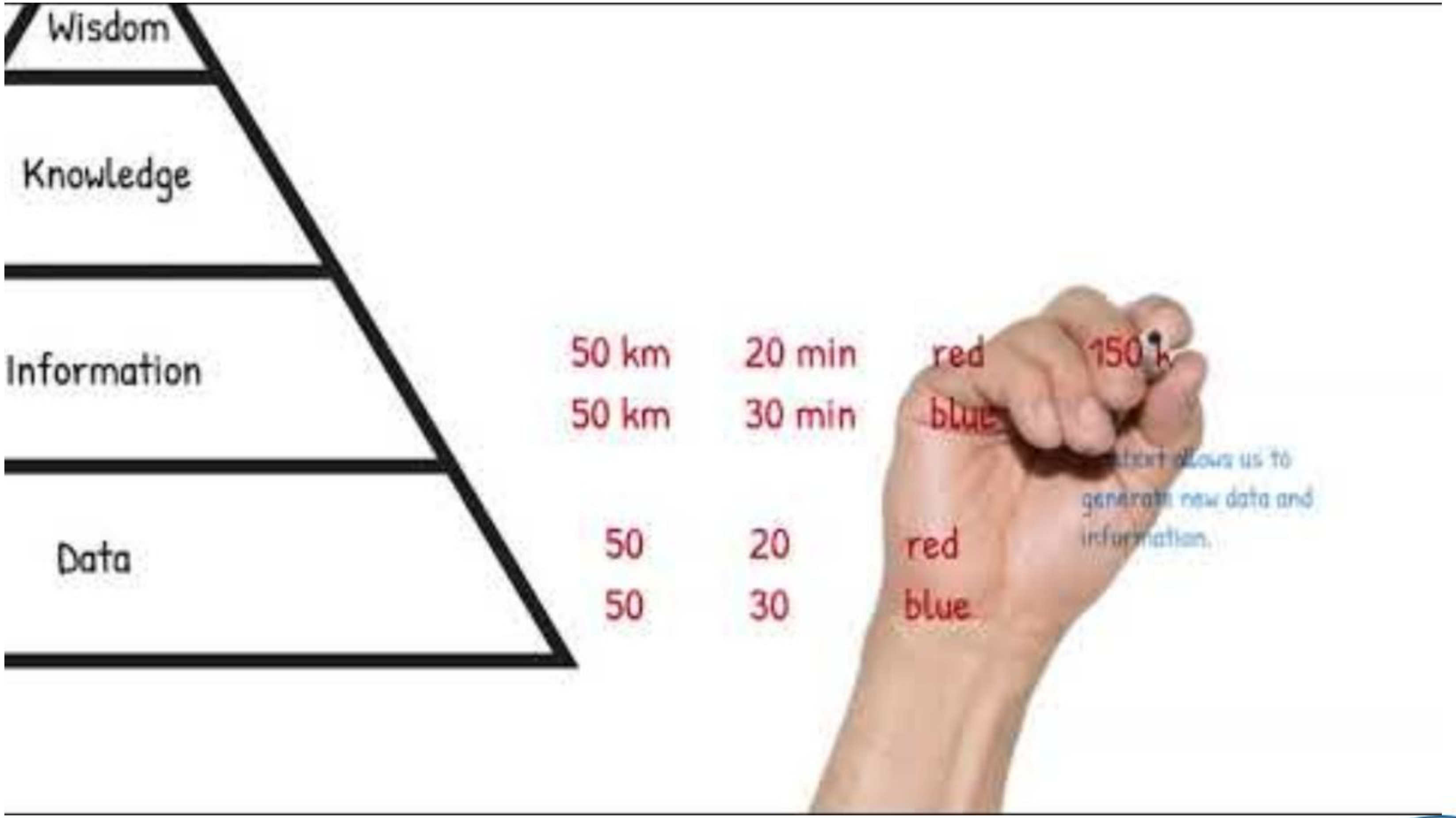
- **Wisdom**: Knowledge applied in action.

The top of the DIKW hierarchy - Guiding **Wisdom**

The DIKW Pyramid



The DIKW Pyramid demo



Part III
Understanding Data

Understanding Data

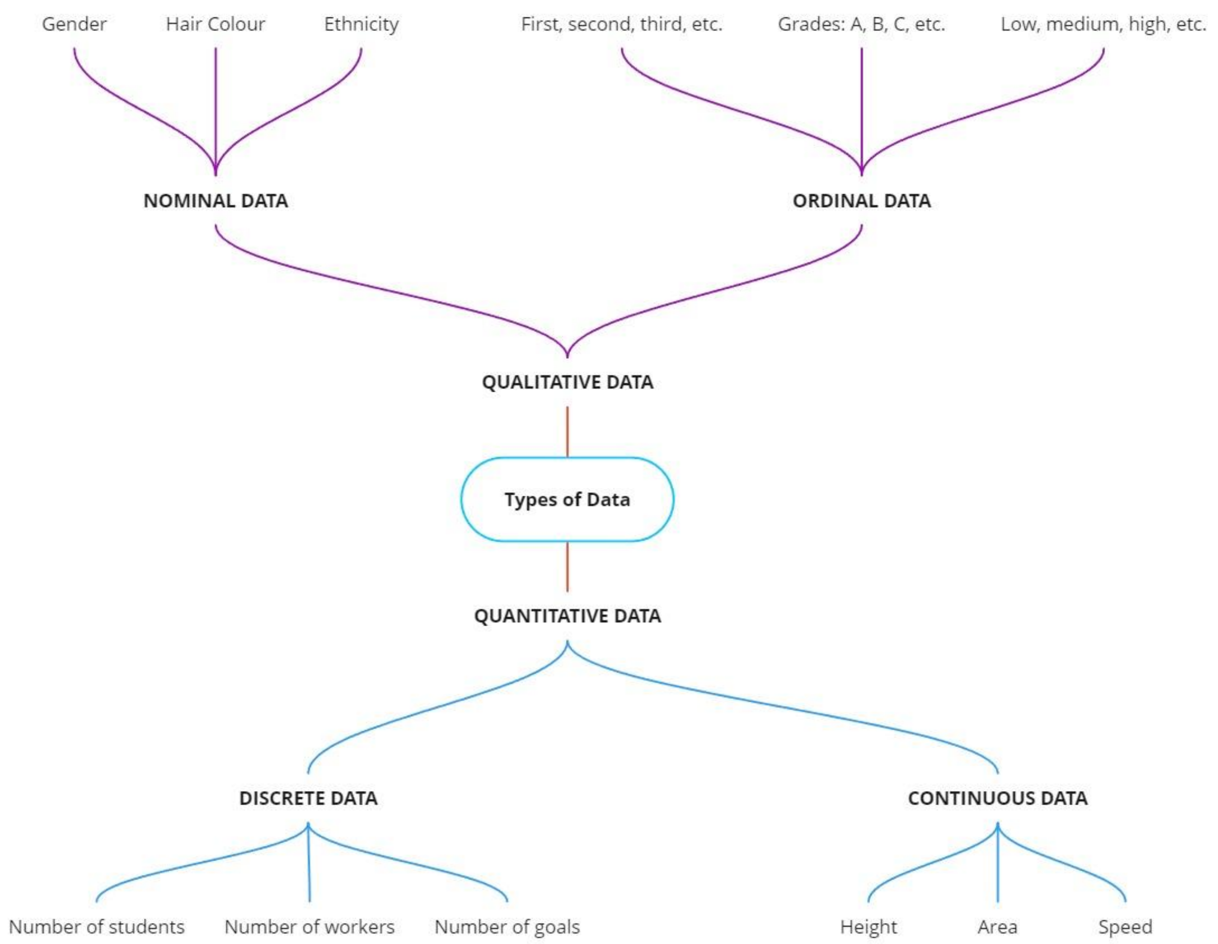
What is Data?

- Understanding data:
  - Data are individual facts, statistics, or items of information. They can also be considered raw values or measures.

Types of Data

- Data can be classified by its values.
- A data type is the classification of the value of the data. It is usually quantitative (numbers) or qualitative (letters or text) in nature.
- Meta-data is a type of data that describes other data.

Types of Data



Types of Data Structures

- Data can have structure.
- How data is organized is its data structure.
- Structured – 2-dimensional tabular data
- Unstructured – variable or dynamic size and format with no pre-set form
- Semi-structured – unstructured data that has some pre-set form

Spreadsheets and databases are typically used to manage structured data.

Data

- Facts or observations about people, places, things, and events.
- Used to be only numbers, letters, and symbols, but now also includes:
  - Audio,
  - Music,
  - Photographs, and
  - Video.
- Two ways to view data:
  - Physical view focuses on the actual format and location of data.
  - Logical view focuses on the meaning, content, and context of the data.

Part IV
Organizing Data in Files

Organizing Data in Files

Organizing Data in a Traditional File Environment

File organization terms and concepts:

- **Bit**: Smallest unit of data; **Bi**nary digi**t** (0, 1)
- **Byte**: Group of bits that represents a single character
- **Field**: Group of words or a complete number
- **Record**: Group of related fields
- **File**: Group of records of same type

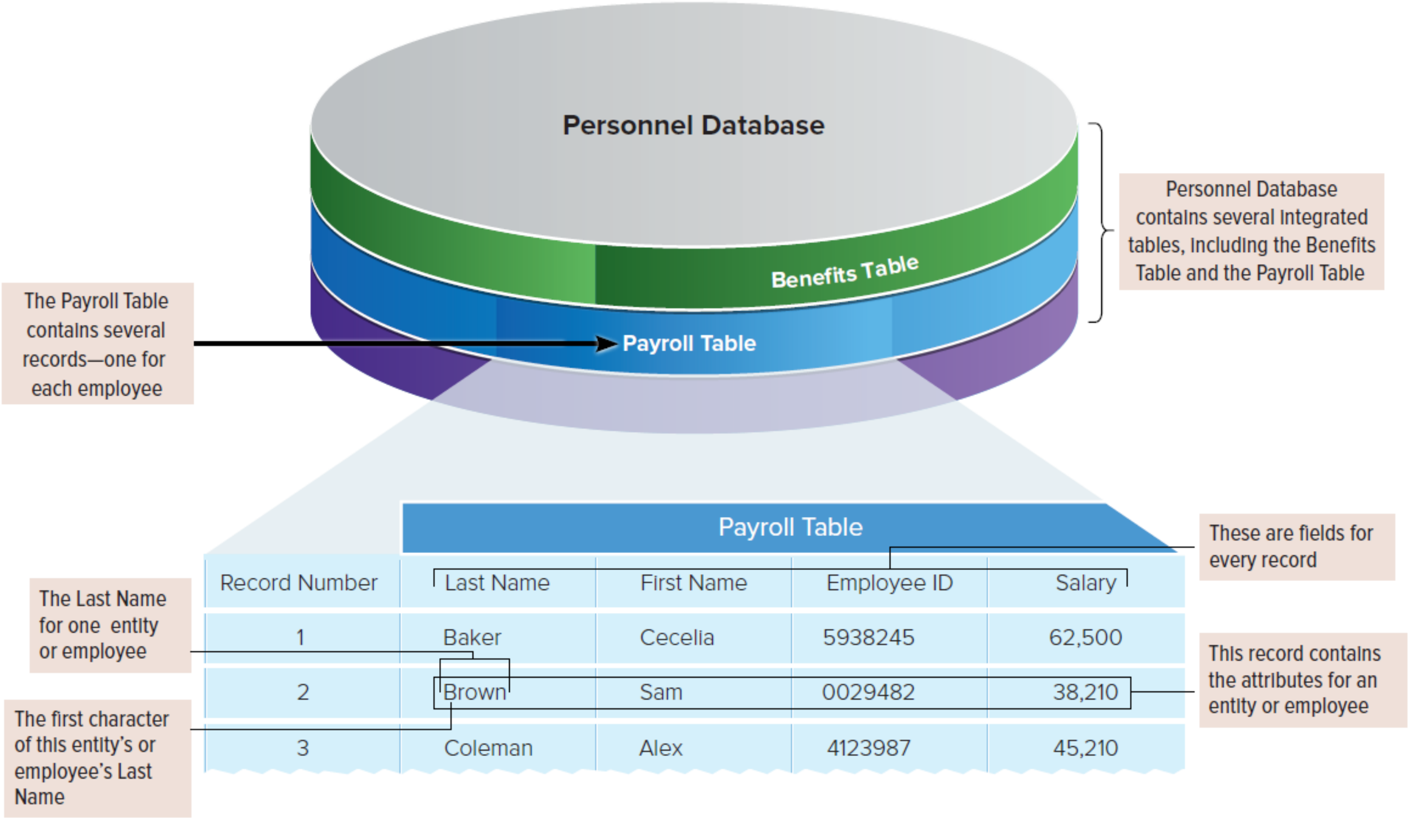
Organizing Data in a Traditional File Environment

- File organization terms and concepts
  - **Database**: Group of related files
  - **Entity**: Person, place, thing, event about which information is maintained
  - **Attribute**: Description of a particular entity
  - **Key field**: Identifier field used to retrieve, update, sort a record

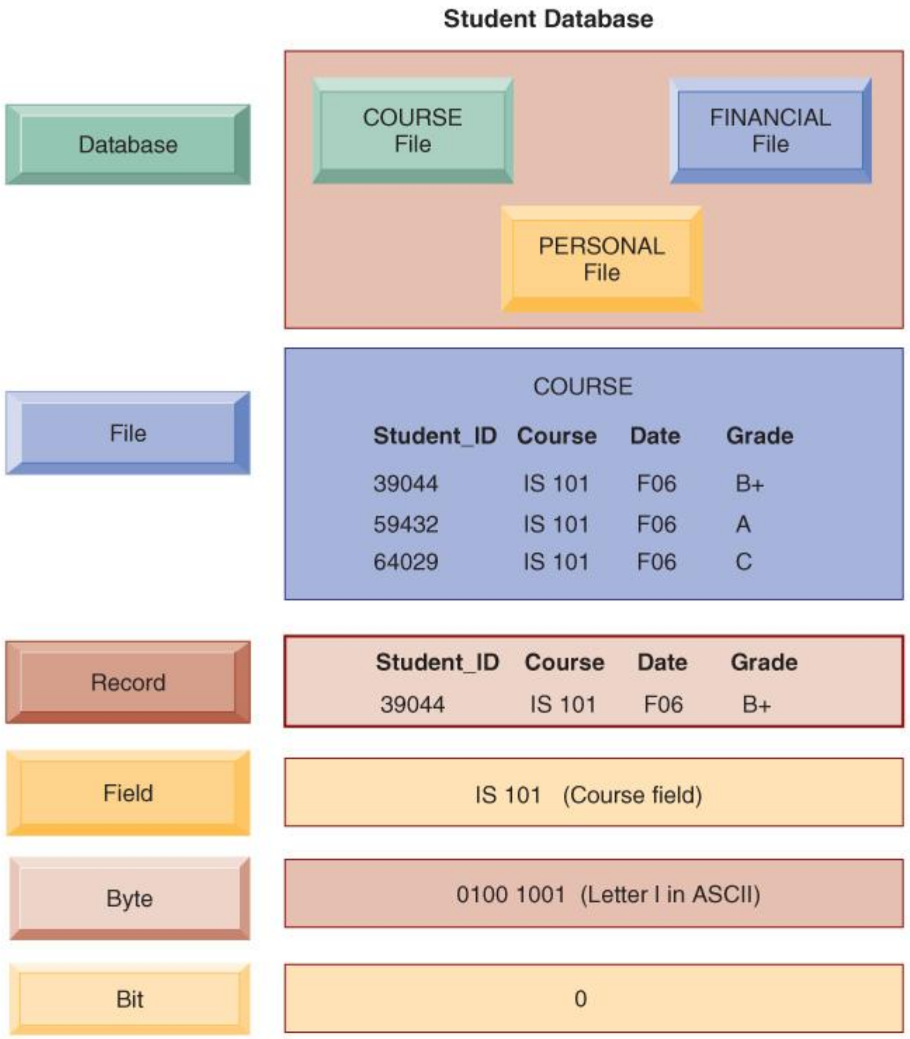
Data Organization

- **Logical view:**
  - Character
  - Field
  - Record
  - Table
  - Database

- **Key Field or Primary Key:**
  - Unique identifier



Student Database



Part VI
Problems in File Environments

Problems in File Environments

Problems with the Traditional File Environment

- Data redundancy and inconsistency
- Program-data dependence
- Lack of flexibility
- Poor security
- Lack of data sharing and availability

Data Redundancy and Inconsistency

- **Data redundancy:** The presence of duplicate data in multiple data files so that the same data are stored in more than one place or location.

- **Data inconsistency**: The same attribute may have different values.

Program – Data Dependency

- The coupling of data stored in files and the specific programs required to update and maintain those files such that changes in programs require changes to the data

Lack of Flexibility

- A traditional file system can deliver routine scheduled reports after extensive programming efforts, but it cannot deliver ad-hoc reports or respond to unanticipated information requirements in a timely fashion

Security and Data Sharing

- Poor security
  - Management may have no knowledge of who is accessing or making changes to the organization’s data.
- Lack of data sharing and availability
  - Information cannot flow freely across different functional areas or different parts of the organization.

Part VIII
About Databases

Introduction

- Like a library, secondary storage is designed to store information.
- End users need to understand.
    - How information is organized in fields, records, tables and databases.
    - The different types of databases and structures.
- Competent end users need to be able to find information that is stored in databases.

Databases Overview

- **Collection of integrated data:**
  - Logically related files and records.
- **Databases address:**
  - Data redundancy – same information in multiple files.
  - Data integrity – accurate updating of files.
- **Advantages to having databases:**
  - Sharing – between departments of an organization.
  - Security – limited access.
  - Less data redundancy – decrease unnecessary duplication.
  - Data integrity – reduce likelihood of inaccurate data.

Database Management

- **Database Management System (D B M S):**
  - Software that enables users to create, modify, and gain access to data.
  - Software made up of:
    - D B M S engine – bridge between the logical view of data and the physical.
    - Data definition subsystem – defines the logical structure by using:
      - Data dictionary or schema.
      - Contains a description of the structure of data.
- **Data manipulation subsystem – tools for maintaining and analyzing data.**
  - Data maintenance
  - S Q L (Structured query language)

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DBMS

- Data manipulation subsystem provides tools for maintaining and analyzing data.

Data Maintenance – maintaining data.

Analysis Tools used to view parts of the data.

  - Query-by-example (QBE).
  - Structured query language (SQL).

- Application generation subsystem provides tools to create data entry.
- Data administration subsystem helps manage the overall database.
    - Database Administrators (DBA’s) administer the database.
    - Processing rights to determine who has access to the databases.

DBMS Structure

- DBMS Programs are designed to work with data that is logically structured or arranged in a particular way.

Database model:

- Model-defined rules and standards for data in a database.

Organization	Description
Hierarchical	Data structured in nodes organized like an upside-down tree; each parent node can have only one parent
Network	Like hierarchical except that each child can have several parents
Relational	Data stored in tables consisting of rows and columns
Multidimensional	Data stored in data cubes with three or more dimensions
Object-oriented	Organizes data using classes, objects, attributes, and methods

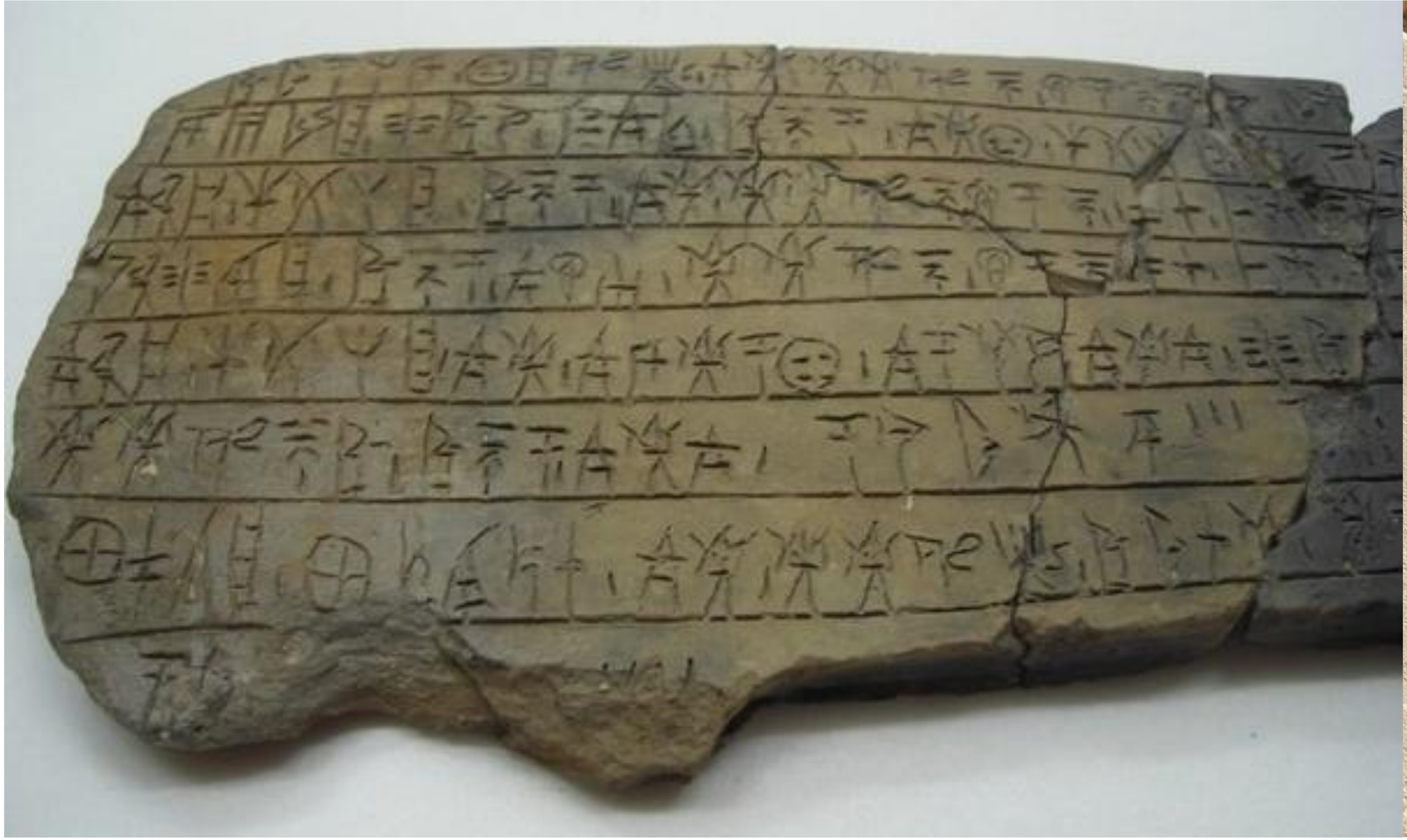
The **Database Approach** to Data Management

- Database management systems solve the problems of traditional file environment:
  - Users have different access to data or views (sharing and security)
  - Relational DBMS create associations between data (addresses redundancy and inconsistency)
  - Operations of a relational DBMS provide standardized interfaces and commands(enables flexibility, program independence)
  - Hierarchical and network DBMS organize data differently (enables flexibility)
  - Object-oriented DBMS organize data differently (enables flexibility)

Part IX
Types of Databases

A Brief History of Databases

Keeping data records …



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History of Databases (1960-2020)

1960s

Databases were coined, two popular data models: Network (Eg. CODASYL) & Hierarchical (Eg. IMS)

1970s

Relational databases (RDBMS) first appear.

1980s

SQL (Standard Query Language) came into limelight. Commercial success to RDBMS.

1990s

Emergence of Oracle, PowerBuilders, VB and others. Excel/Access became defacto for personal productivity. Non-relational databases emerge.

2000s

Object oriented programming came into picture, new applications were developed and databases continued to grow as Internet became mainstream.

2010s

APIs became mainstream. New age database applications came about, mimicking the ease of use of a spreadsheet. New players emerge in the non-relational database space.

2020...

Rise of low/no-code applications, APIs and easy to use, non-technical databases continue to take on the world and power next generation of application development

stackby.com

A Brief History of Databases

- Read a comprehensive article about databases and their history: https://stackby.com/blog/what-is-a-database/

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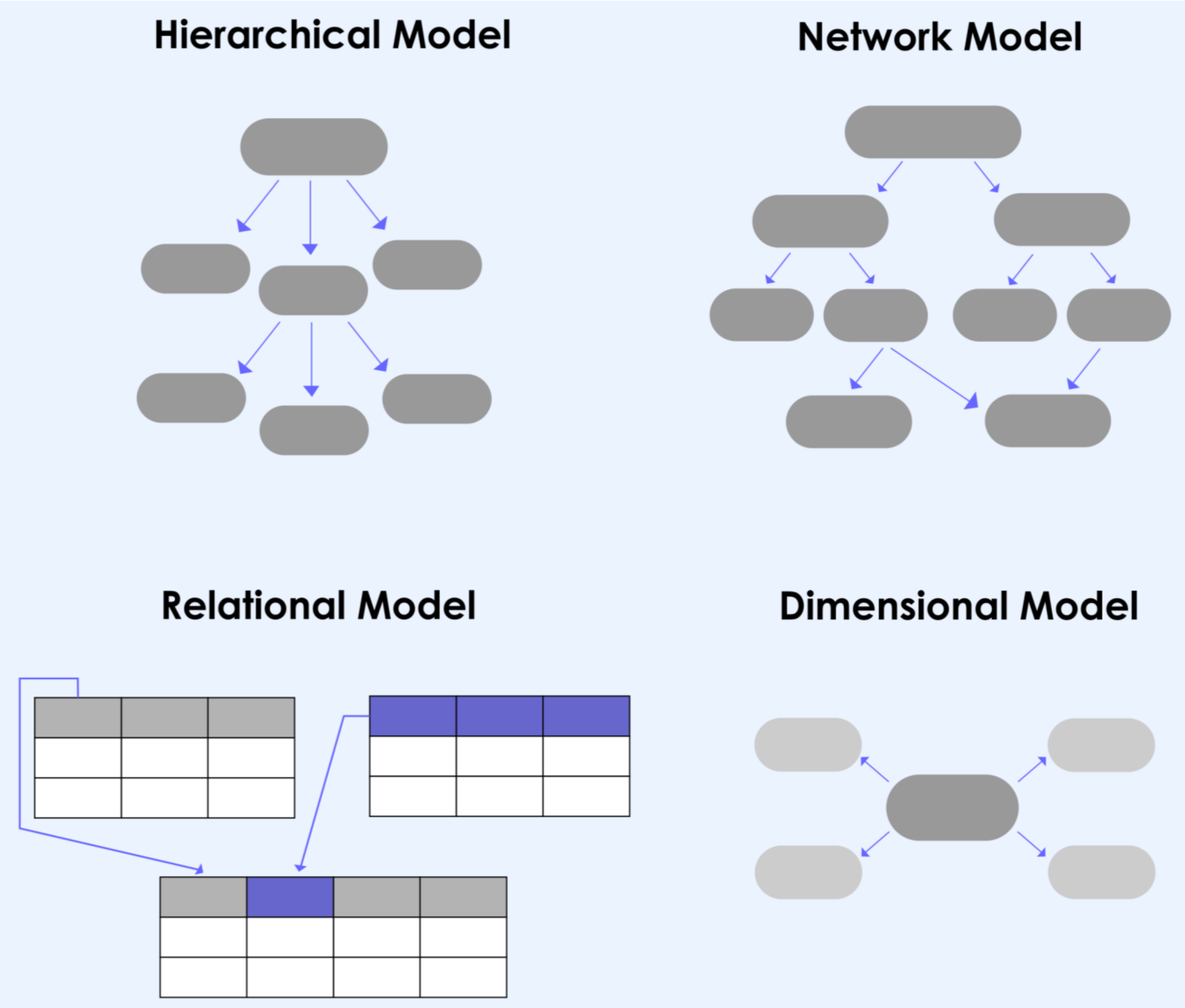
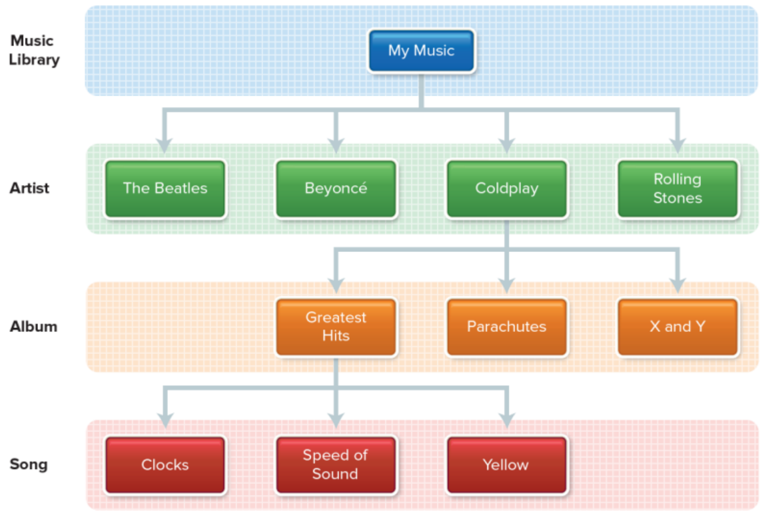
Various Database Types

Database Models

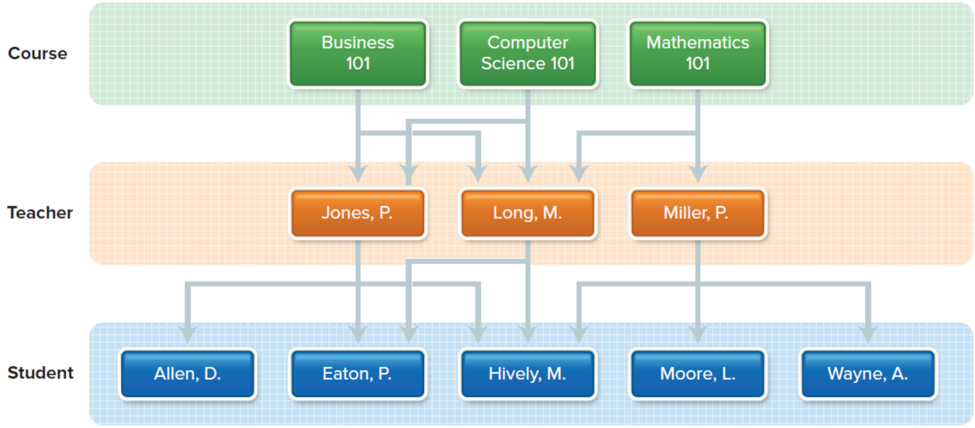


Hierarchical Database



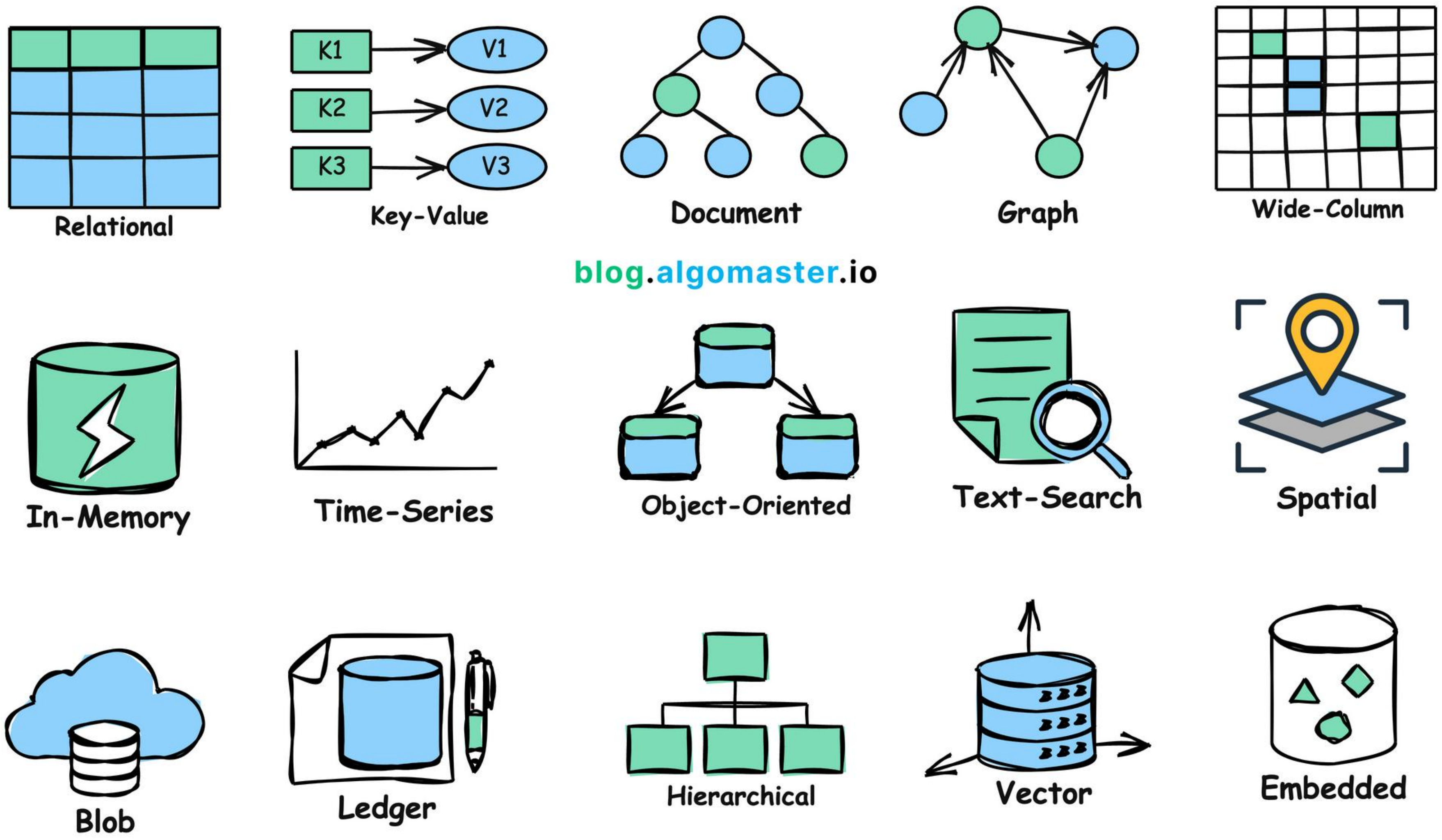
- Fields or records structured in nodes.
- Nodes:
  Points connected like branches of an upside-down tree.
  - One parent per node.
  - Parent can have several child nodes.
  - One-to-many relationship.
- Major concern is that if your parent node is deleted then so are all subordinate child nodes.

Network Database



- Has a hierarchical node arrangement.
- Each child node may have more than one parent node.
  - many-to-many relationship.
- Additional connections between parent and child are *Pointers.*
- Nodes can be reached through multiple paths.

Database Types



Database Types (Performance)

1 RELATIONAL DATABASES

- Tabular data structure
- Uses SQL
- ACID properties
- Suitable for structured data, complex queries, transactions
- Examples: MySQL, PostgreSQL, Oracle Database

2 NOSQL DATABASES

- Unstructured data
- Flexible schema
- Scalable for big data
- Types: document, key-value, columnar, graph
- Examples: MongoDB, Cassandra, Redis, Neo4j

3 GRAPH DATABASES

- Stores data as nodes, edges, properties
- Ideal for complex relationship queries
- Uses high-level query languages like Cypher
- Examples: Neo4j, Amazon Neptune

4 TIME-SERIES DATABASES

- For time-stamped data
- Optimized for insertion and retrieval
- Suitable for IoT and monitoring
- Examples: InfluxDB, Prometheus

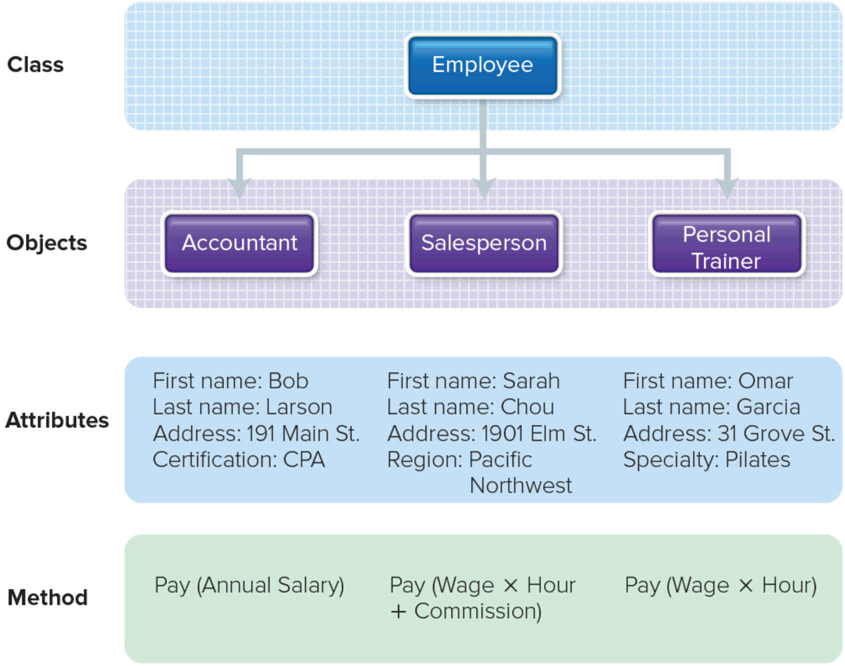
5 DOCUMENT DATABASES

- Uses flexible, JSON-like documents
- Supports dynamic schemas
- Suitable for content management, e-commerce
- Examples: MongoDB, Couchbase

6 In-Memory Databases

- Main memory storage for speed
- For real-time analytics, caching
- Examples: Redis, Apache Ignite

Object-Oriented Database



Store data as well as instructions to manipulate data.

- Organize data using:
  - Classes – general definitions.
  - Objects – specific instances of class containing data and instructions to manipulate the data.
  - Attributes – data fields an object possesses.
  - Methods – instructions for retrieving or manipulating attribute values.

Multidimensional Database

- A variation and an extension of the relational model.
- Data cube.
- Extension of the two-dimensional data model to include additional or multiple dimensions.
- Good for representing complex relationships.
- Advantages over relational databases.
  - Conceptualization provides users with an intuitive model in which complex data and relationships can be conceptualized.
  - Processing speed for analyzing and querying a large multidimensional database is faster.

Non-Relational Databases and Databases in the Cloud

- Non-Relational Databases
  - NoSQL:
    - NO SQL, or
    - Not Only SQL

    is a non-relational database that uses a more flexible data model and are designed for managing large datasets
  - Many different kinds, including Oracle's NoSQL and Amazon's SimpleDB
  - Examples of NoSQL content could be document repository
  - NoSQL content is generally searched using programming languages/syntax and is an ideal candidate for Natural Language Processing

- Cloud Databases
  - Private clouds consolidate servers, storage, operating systems, databases onto a shared hardware and software architecture
  - Easier to store Petabytes of data centrally on the cloud than move large datasets around to remote locations for analysis

Types of Databases

Commercial databases:

- Oracle, Microsoft SQL Server, IBM DB2, etc.

Open-Source databases:

- PostgreSQL, MariaDB, MySQL, Apache Derby, etc.

Type	Description
Individual	Integrated files used by just one person.
Company	Common operational or commonly used files shared in an organization.
Distributed	Database spread geographically and accessed using database server.
Commercial	Information utilities or data available to users on a wide range of topics.

Part X
The Relational Database Management System

What is a Relational Database?

- A relational database is a collection of information that organizes data in predefined relationships where data is stored in one or more tables (or "relations")
- Tables contain *columns* and *rows*, making it easy to see and understand how different data structures relate to each other
- Relationships are a logical connection between different tables, established on the basis of interaction among these tables

The Relational Database Model

- Developed by E.F. Codd from IBM in the 1970s.
- The relational database model allows any table to be related to another table using a common attribute.
- Instead of using hierarchical structures to organize data, Codd proposed a shift to using a data model.
- Data is stored, accessed, and related in tables without reorganizing the tables that contain them.

The Relational Database Model

- Think of the relational database as a collection of spreadsheet files that help businesses organize, manage, and relate data.
- In the relational database model, each “spreadsheet” is a TABLE that stores information, represented as COLUMNS (attributes) and ROWS (records or tuples).
- Attributes (columns) specify a data type.
- Each record (or row) contains the value of that specific data type.
- All tables in a relational database have an attribute known as the primary key, which is a unique identifier of a row.
- Each row can be used to create a relationship between different tables using a foreign key—a reference to a primary key of another existing table.

Relational Database

- A more flexible type where there are no access paths down a hierarchy.
- Data stored in table called a relation.
- Tables linked via a common data item.
- Tables consist of rows and columns.

Vehicle Owner Table

License Number	Last Name	First Name	Street	City	State	Zip
JF1987	Aaron	Linda	132 Park Lane	San Jose	CA	95127
D12372	Abar	John	139 Oak St.	Lakeville	CA	94128
LK3457	Abell	Jack	9587 State St.	Stone	CA	95201
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Key fields related

Key fields related

Vehicle Table

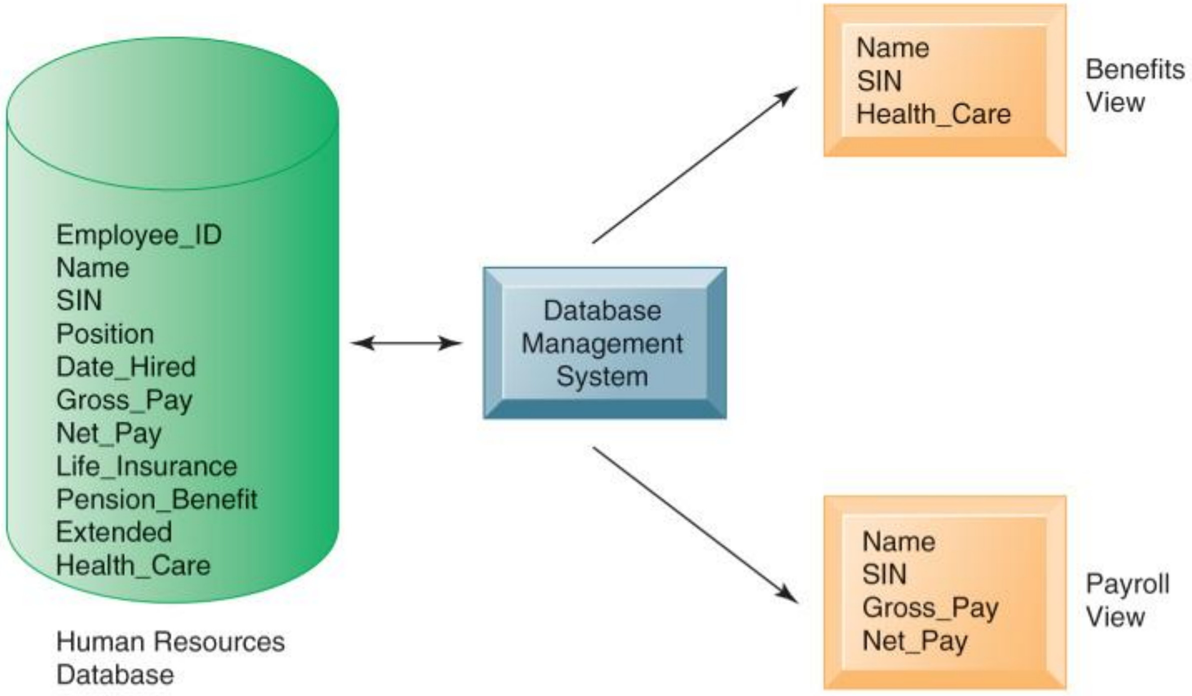
License Number	Vehicle
LK3457	1997 Ford Explorer
BD1395	2009 Toyota Camry
JR1321	2008 Chevy Suburban
⋮	⋮

Outstanding Citation Table

License Number	Citation Code
SX9847	Speed
DPL415	Park
LK3457	Speed
⋮	⋮

Relational Database Management System (RDBMS) Example

A single human resources database provides many different views of data, depending on the information requirements of the user. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.



Relational DBMS

- Represents data as two-dimensional tables called relations
- Relates data across tables based on common data element
- Examples:
  - Access, DB2, Oracle, MS SQL Server, etc.
  - PostgreSQL, MariaDB, MySQL, SQLite, Apache Derby, etc.

Tables in Relational Databases

FIGURE 6-4 Relational database tables.

SUPPLIER

Columns (Attributes, Fields)

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_Province	Supplier_PC
8259	CBM Inc.	74 5th Avenue	Saint John	NB	E2M 5T3
8261	B. R. Molds	1277 Gandolly Street	Calgary	AB	T2E 0B3
8263	Jackson Composites	8233 Micklin Street	Hamilton	ON	L9H 7M4
8444	Bryant Corporation	4315 Mill Drive	Richmond	BC	V7E 3M1

Rows (Records, Tuples)

Key Field (Primary Key)

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door moulding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Primary Key

Foreign Key

A relational database organizes data in the form of two-dimensional tables. Illustrated here are tables for the entities SUPPLIER and PART showing how they represent each entity and its attributes. Supplier_Number is a primary key for the SUPPLIER table and a foreign key for the PART table.

Operations of a Relational DBMS

- **SELECT:** Creates subset of rows that meet specific criteria
- **JOIN:** Combines relational tables to provide users with information
- **PROJECT:** Enables users to create new tables containing only relevant information

SELECT – PROJECT – JOIN

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door moulding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Select Part_Number = 137 or 150

SUPPLIER

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_Province	Supplier_PC
8259	CBM Inc.	74 5th Avenue	Saint John	NB	E2M 5T3
8261	B. R. Molds	1277 Gandolly Street	Calgary	AB	T2E 0B3
8263	Jackson Components	8233 Micklin Street	Hamilton	ON	L9H 7M4
8444	Bryant Corporation	4315 Mill Drive	Richmond	BC	V7E 3M1

Join by Supplier_Number

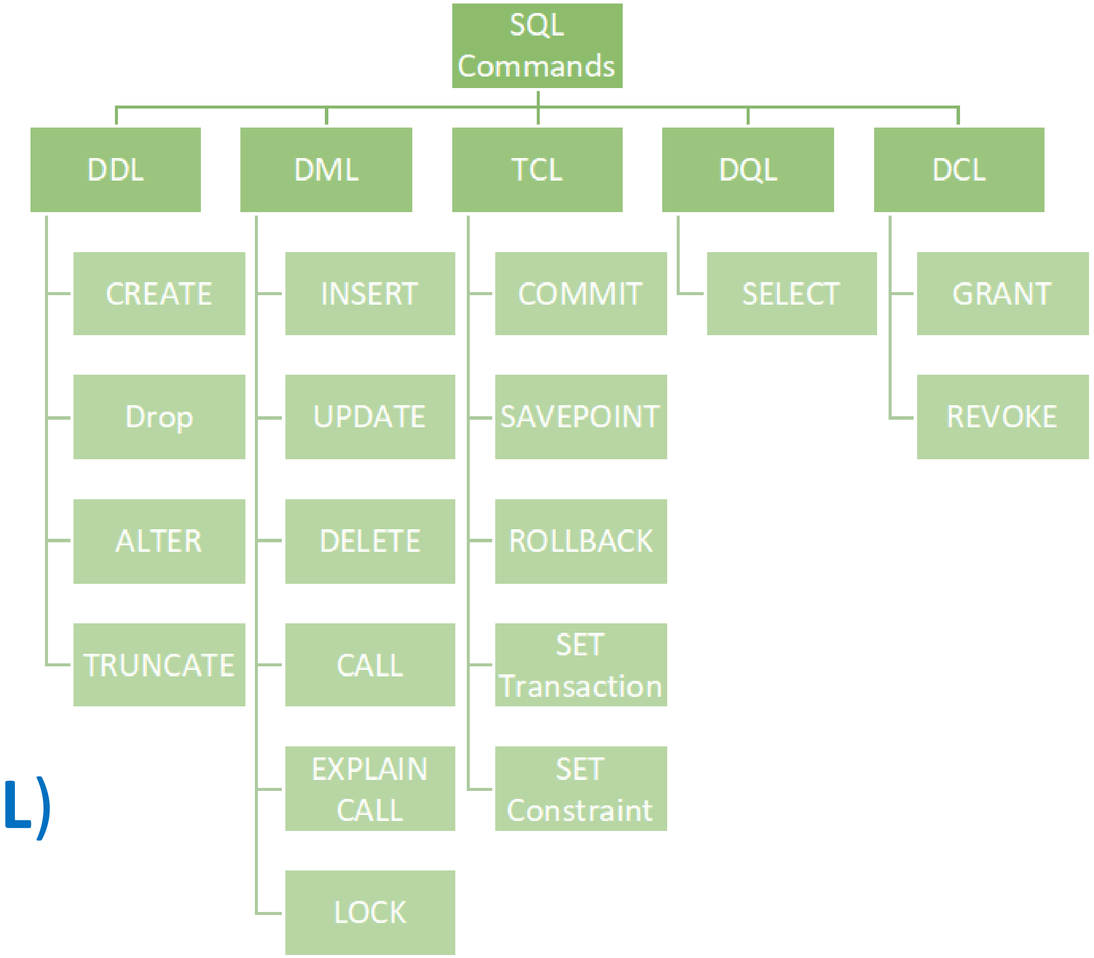
Part_Number	Part_Name	Supplier_Number	Supplier_Name
137	Door latch	8259	CBM Inc.
150	Door moulding	8263	Jackson Components

Project selected columns

The select, join, and project operations enable data from two different tables to be combined and only selected attributes to be displayed.

Managing Database Systems

- **Data Dictionary**
  - Stores definition of data elements and their characteristics
- **Structured Query Language (SQL):**
  - **Data Definition Language (DDL)**
  - Data Query Language (DQL)
  - **Data Manipulation Language (DML)**
  - Data Control Language (DCL)
  - Transaction Control Language (TCL)



Integrity Constraints

- Integrity constraints are rules to be enforced to maintain quality and consistency of data.

- There are four types of integrity constraints in DBMS:
  - Domain Constraint (e.g., restricted data types for fields)
  - Entity Constraint (e.g., No null values for primary key)
  - Referential Integrity Constraint (e.g., Must be valid relationship between two tables)
  - Key Constraint (e.g., No duplicate values for primary key)

NULL Values

- A missing (undefined) value in a field.
- Not to be confused with:
  - 0 … which is a value (48 in ASCII), or
  - Space (‘ ‘), for strings (CHARACTER Data Type).

Geek Fun! Open up Notepad and try entering ASCII values into it. To do this, hold ALT+<ascii value> (but you need to have a numeric keypad).

One byte = 2^8 values or 256 values. They range from 0 ... 255.

https://www.techonthenet.com/ascii/chart.php

https://www.rapidtables.com/code/text/alt-codes.html

Part XI
Wrap-up – What's Next

Summary

Summary

In this lecture we discussed about:

- The DIKW Pyramid
- Understanding Data
- Organizing Data in Files
- Problems in File Environments
- Databases
- Types of Databases
- RDBMS

Next Lecture

Next Lecture

- RDBMS:
  - Data Definition Language (DDL)
  - Data Manipulation Language (DML)
  - JOINS in SQL
- Hands-on to Database Programming in Java

Part XII
Training and resources

Tools

Tools

- https://www.essentialsql.com/sql-tutorial-for-sql-server/
- https://www.w3schools.com/sql/sql_editor.asp
- https://www.programiz.com/sql/online-compiler/
- https://sqliteonline.com/
- https://livesql.oracle.com/apex/f?p=590:1000
- Visual Studio Code
- Notepad++

- DBeaver: https://dbeaver.io/
- SqlDBM: https://app.sqldbm.com/
- https://dbmstools.com/
- Oracle: https://www.oracle.com/database/sqldeveloper/technologies/sql-data-modeler/

Tools

- https://www.essentialsql.com/sql-tutorial-for-sql-server/
- https://www.w3schools.com/sql/sql_editor.asp
- https://www.programiz.com/sql/online-compiler/
- https://sqliteonline.com/
- https://livesql.oracle.com/apex/f?
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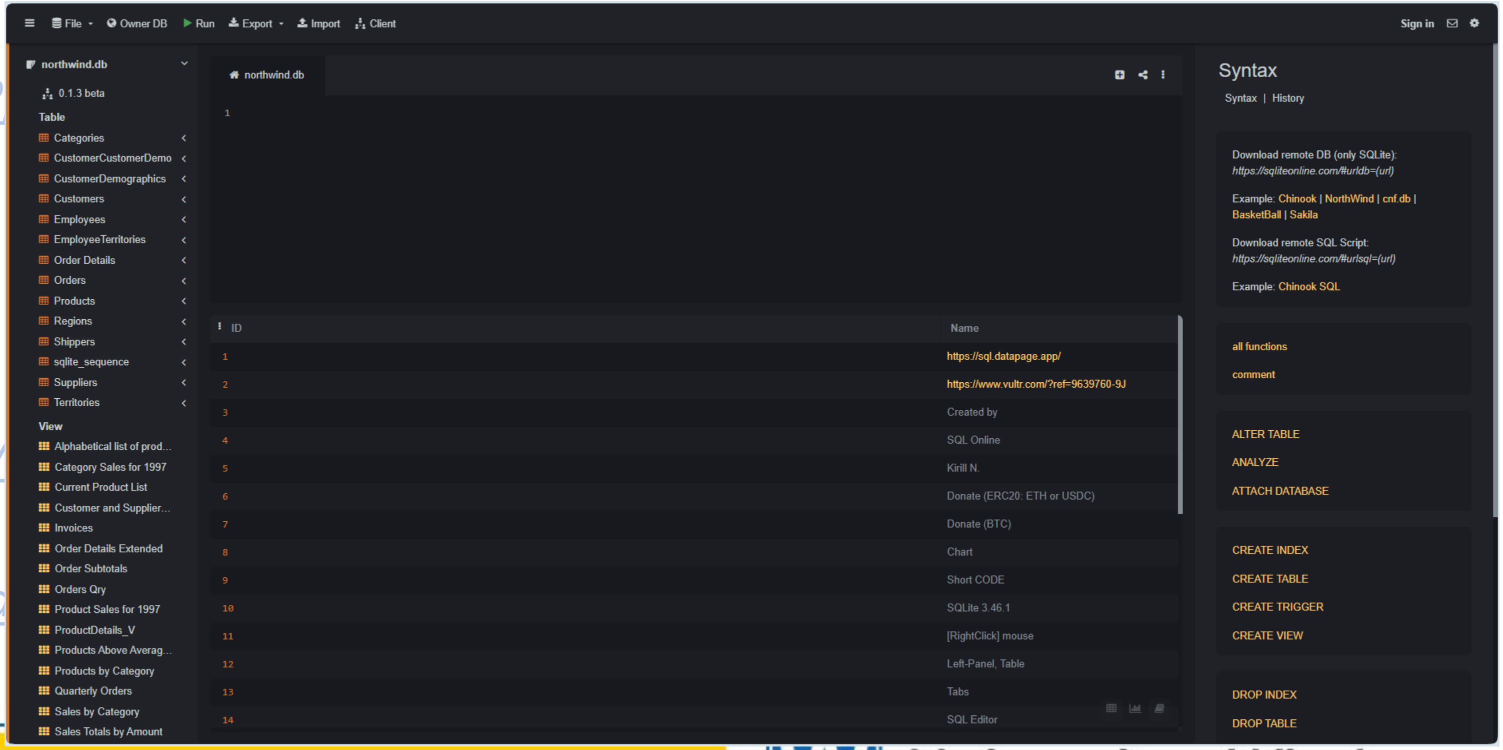
- SqlDBM: https://app.sqldbm.com/
- https://dbmstools.com/
- Oracle: https://www.oracle.com/c



Training resources

Training Resources

- Datacamp: https://app.datacamp.com/learn/courses/introduction-to-sql
- https://www.essentialsql.com/
- https://www.w3schools.com/sql/
- https://www.khanacademy.org/computing/computer-programming/sql/sql-basics/v/welcome-to-sql
- https://www.learnsqlonline.org/
- https://sqlbolt.com/
- https://sqlzoo.net/wiki/SQL_Tutorial
- https://www.codecademy.com/resources/docs/sql
- https://www.freecodecamp.org/
- …

Thank you!

Questions?

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